EXP NO: 18

ECHO CLIENT - SERVER USING TCP/UDP SOCKETS

AIM:

To implement an echo client-server using TCP/UDP sockets.

CODE (TCP):

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET,
            socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] connected listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
        print("[SERVER] connection closed by client")
        break
        print(f"[SERVER] received: {data}")
        conn.send(f"server received: {data}".encode())
    conn.close()
    server.close()
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_
    STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] connected to server")
    while True:
        msg = input("Enter message or ('quit'
        to exist):")
```

```
            client.send(msg.encode())
            if msg.lower() == 'quit':
                break
            response = client.recv(1024).decode()
            print(f" [SERVER RESPONSE] {response}")
        client.close()
if __name__ == "__main__":
    server_thread = threading.Thread(target=start_
server_thread.start()
    start_client()
```

OUTPUT :

```
Enter message (or 'quit' to exit): hello
Receive hello
server received :hello
Enter message (or 'quit' to exit : quit' connection
closed by client
```

CODE : UDP

```
import socket
import threading
def udp_server():
    S = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    S.bind(("127.0.0.1", 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = S.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended the connection")
            break
        print(f"[SERVER] Received from {addr} : {msg}")
        S.sendto(data, addr)
    S.close()
    print("[SERVER] stopped.")
```

```
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] closing connection")
            break
        data, _ = c.recvfrom(1024)
        print("[CLIENT] stopped.")

if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

OUTPUT:

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1: 12000 hi
Received from ('127.0.0.1', 49670) : hi
Echo from server : hi
Enter message (or 'quit' to stop) : hello
Received from ('127.0.0.1', 49670) : hello
Echo from server : hello
Enter message (or 'quit' to stop) : how are you
Received from ('127.0.0.1', 49670) : how are you
Echo from server : how are you
Enter message (or 'quit' to stop) : quit
[CLIENT] closing connection.
[SERVER] Client ('127.0.0.1', 49670) ended the connection.
```

RESULT:

Thus, an echo-client server is implemented using TCP and UDP sockets